Arpan Ghosh

Machine Learning · Data Science · AI Engineer

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SUMMARY

M.Tech CSE student at Tripura University building end-to-end AI products: a face-and-voice attendance system (Attendly), an LLM contract analyzer (NibbleContract AI) and a real-time pose-based gym coach (Movexa), all deployed live. IEEE-published co-author in computer vision. Seeking remote ML, Data Science and AI Engineer internships.

EDUCATION

M.Tech (CSE), Tripura University — 2026 – 2028 · Pursuing

BE (CSE), Chandigarh University — 2021 – 2025 · CGPA 7.4

Class XII, WBCHSE · **Class X**, WBBSE — 93.4% · 87%

STACK

ML & statistics: scikit-learn, XGBoost, SVM, regression, ensembles, clustering, PCA, SciPy

Data & viz: SQL, pandas, NumPy, Matplotlib, Seaborn

LLMs & APIs: RAG, LangChain, ChromaDB, Hugging Face, Groq & OpenAI APIs, REST APIs, Tavily, RDAP

Deep learning: PyTorch, TensorFlow, ANN / FNN, CNN, RNN, U-Net

Vision & audio: OpenCV, dlib, face & speaker embeddings, Resemblyzer, librosa

Tools & infra: Python, Flask, Streamlit, Git, Docker, Render, Supabase, Jupyter

PROJECTS

NibbleContract AI · Contract & Internship Authenticity Checker

Live demo · GitHub

Python · Flask · Groq LLM · OpenAI-compatible API · pypdf · RDAP · Tavily

- Turns internship offers, rental and loan agreements (PDF or text) into a plain-language risk report: ranked red flags, deadlines, money terms and questions to ask before signing.
- Explainable 0–100 fake-internship trust score: the LLM only extracts evidence, fixed rules score it, unverified quotes are dropped; RDAP domain-age, SSRF-safe site checks and Tavily web + Reddit review search add signals.
- Tested on 30 real documents: 90% of red flags caught, 93% fake-offer detection accuracy.

Attendly · AI Attendance with Face & Voice Recognition

Live demo · GitHub

Python · Streamlit · dlib ResNet embeddings · scikit-learn SVC · Resemblyzer · librosa · SQL

- Marks a whole class from one group photo: 128-d face embeddings plus an SVC classifier identify each student, and a distance threshold rejects unknown faces.
- Voice roll-call matches each speaker by embedding similarity; teacher and student portals with secure logins.
- 95% face-match accuracy across 25 students, 2% false-accept and 4% false-reject rate.

Movexa · AI Gym Coach & Live Form Tracker

Live demo · GitHub

Python · Pose estimation (33 landmarks) · RAG · LLM API · Docker · WebRTC

- Builds a personalised gym or home workout plan (sets, reps, calorie and protein targets) from body stats, goal and equipment, grounded by RAG over exercise-science guidance and adapted around injuries.
- Live form coach: camera pose estimation measures joint angles, counts reps and sets, and gives on-screen and spoken cues when form breaks; 54 exercises, 95% rep-count accuracy, ~30 ms per frame.

Machine Learning & Deep Learning Research

ML repo · DL repo

scikit-learn · XGBoost · PyTorch · TensorFlow / Keras · pandas · Matplotlib · Seaborn

- **Machine learning:** supervised and unsupervised learning on real datasets, from regression, SVM and ensembles to clustering, PCA and anomaly detection, compared through visualization.
- **Deep learning:** ANN / FNN, CNN and RNN models in PyTorch and TensorFlow for tabular, image and text tasks, plus a RAG pipeline with LangChain and ChromaDB.
- **Minor projects:** CreditWise Loan Approval · House Prices · Insurance Charges · Employee Turnover · Heart Disease · Titanic · Shop Smart · Thyroid Anomaly · Date Fruit · Power Plant · Handwritten Characters · ASL Sign Language

PUBLICATION

A login confirmation mechanism for meeting platforms with an automatic face acknowledgment system · Co-author · IEEE CCICT 2024 · LBPH & Haar-cascade face recognition